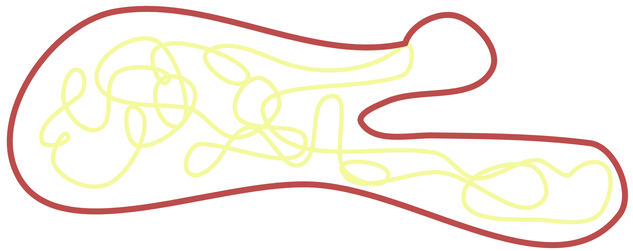
NIGnets and Neural ODEs for Non-Self-Intersecting Geometry

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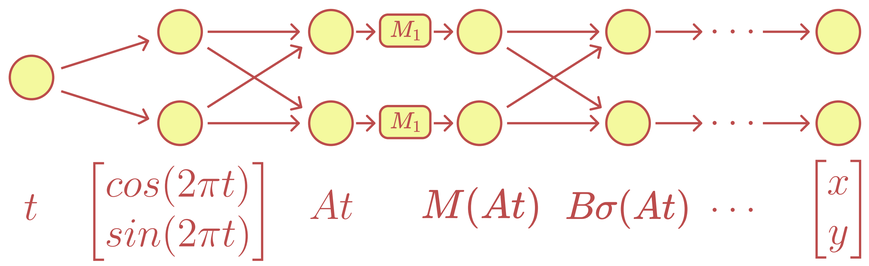
The Problem: Self-Intersection

A large class of shape optimization problems require representing simple closed curves, i.e. closed curves that do not self-intersect. Self-intersecting shapes can lead to various issues during shape optimization. One such issue is Design Space Blow-Up. Imagine flow going left → right over the shape below. It never interacts with the internal representation making it wasteful. This leads to optimizers searching through a design space bigger than they should.



NIGnets and Neural ODEs

NIGnets: Neural Injective Geometry networks are a new neural architecture we developed, the core architecture of which looks as follows:



Neural ODEs: Another method is to frame the shape representation as an Initial Value Problem ODE. We start with an initial shape, a circle, which is modified by an ODE (evolution function defined by a neural network) to represent a final target shape.

$$\frac{d\mathbf{h}(t)}{dt} = f(\mathbf{h}(t), t, \theta)$$

Limitations and Future Work

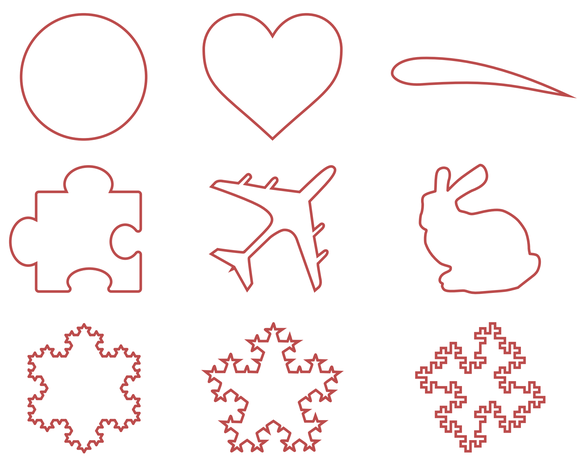
1. Invertible Residual Networks: Use iResNets for the discrete deformation version of Neural ODEs.
2. Extend the comparison to 3D to analyze the effect of curse of dimensionality.
3. Compare performance of NIGnets and Neural ODEs on real world shape optimization tasks.

References

1. Chen, Ricky TQ, et al. "Neural ordinary differential equations." Advances in neural information processing systems 31 (2018).
2. Aalok, Atharva, et al. NIGnets, https://github.com/atharvaaalok/NIGnets (2025).
3. Igel, Christian. "Smooth min-max monotonic networks." arXiv preprint arXiv:2306.01147 (2023).

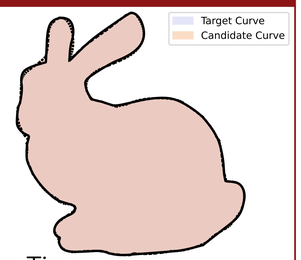
Dataset

We develop a dataset of SVG shapes that can be sampled at arbitrary resolution. The dataset consists of: low, intermediate and high complexity shapes.



Results

We perform a wide range of experiments comparing the performance of NIGnets and Neural ODEs based on: 1) Accuracy of Representation, 2) Training Time and 3) Inference Time. Throughout our experiments we use Adam as our optimizer with learning rate and total epochs of 0.1, 10000 for NIGnets and 0.01, 1000 for Neural ODEs because the former is 10x faster. For a fair comparison we use the same number of parameters, 2800, in both our representation methods. Since Neural ODEs fail to work well with MSE Loss, we use Chamfer Loss as our objective for shape fitting. Accuracy of Representation is reported after averaging over 10 runs and Inference Times are averaged over 100 runs.



Accuracy of Representation

Shape	NIGnet	Neural ODE
circle	1.41e-5	1.51e-5
square	4.55e-5	5.20e-5
airfoil	4.60e-6	2.17e-5
heart	5.07e-5	2.61e-5
stanford bunny	3.69e-4	5.35e-4
hand	2.23e-3	1.10e-2
puzzle piece	6.87e-4	3.59e-4
airplane	1.38e-3	1.13e-3

Training Time

Shape	NIGnet	Neural ODE
circle	102.5s	40.1s
square	99.2s	57.5s
airfoil	100.3s	130.2s
heart	100.9s	52.3s
stanford bunny	105.5s	88.1s
hand	108.3s	125.0s
puzzle piece	105.6s	99.4s
airplane	105.1s	127.2s

Inference Time

Num Pts on Shape	NIGnet	Neural ODE
10	0.00085s	0.01552s
100	0.00142s	0.02059s
1000	0.00336s	0.03674s
10000	0.01705s	0.13668s
100000	0.11853s	0.78579s
1000000	1.13743s	10.23218s